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Synthesis, crystal structures, and DNA-binding studies of two silver(I) complexes with 1,3-bis(1-ethylbenzimidazol-2-yl)-2-thiapropane

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ABSTRACT

Two new silver(I) complexes, $[Ag(\text{bebt})(\text{methacrylate})]$ (**1**) and $[Ag_2(\text{bebt})_2(\text{pic})_2]$ (**2**) ($\text{bebt} = 1,3\text{-bis}(1\text{-ethylbenzimidazol-2-yl})\text{-2-thiapropane}$, $\text{pic} = \text{picrate}$), have been synthesized and characterized by physico-chemical and spectroscopic methods. Single-crystal X-ray diffraction revealed that the structure of **1** was four-coordinate in a distorted tetrahedral geometry, while in **2**, it may be described as a $2 + 1$ coordinated silver(I) compound with a nearly linear $N\text{-Ag}\text{-N}$ backbone and a weak $\text{Ag}\text{-S}$ interaction. Two ligands are arranged in a face-to-face syn conformation to coordinate to two $Ag(I)$ ions from opposite directions in **2**, generating a U-type double layer structure around the metal ions. Experimental studies of the DNA-binding properties indicated that free bebt and both complexes bind to DNA via intercalation, and the order of the binding affinity is $2 > 1 > \text{bebt}$.

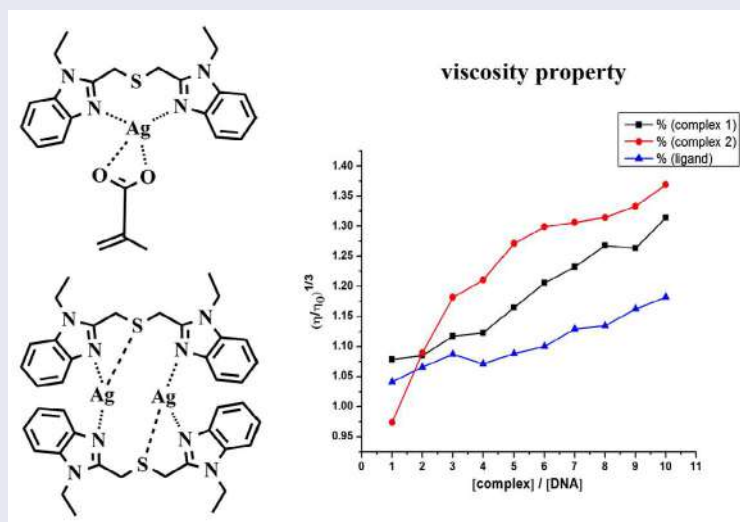
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Benzimidazole; silver(I) complex; crystal structure; DNA-binding property; synthesis



1. Introduction

Bis-benzimidazole and its derivatives are key components in many bioactive compounds of both natural and synthetic origin; bis-benzimidazole ligands, as a class of aromatic N-donor organic linkers, are still less developed [1–3]. Properties of benzimidazole and its derivatives have been studied with broad spectrum of biological activities. The benzimidazole moiety is found in various biologically active compounds having antiviral, antiulcer, antihypertensive, anticancer, and antihistaminic properties [4–8]. Studies on the interaction of transition metal complexes with DNA continue to attract the attention of researchers due to their importance in design and development of synthetic restriction enzymes, chemotherapeutic drugs, and pharmacological studies [8–15]. There has been increasing focus on the binding of small molecules to DNA during the last decades [16–18].

Silver has been used as an antimicrobial agent since antiquity, and it was earlier employed in the purification of drinking water, wine and vinegar, and also Hippocrates remarked on its therapeutic properties [19]. Silver ions and silver compounds are toxic to some bacteria, viruses, algae, and fungi. For that reason, silver-based medicines have been widely used for centuries.

In previous papers, we have investigated a series of V-shaped bis-benzimidazole ligands and their complexes [20, 21]. In this present work, the synthesis and characterization of 1,3-bis(1-ethylbenzimidazol-2-yl)-2-thiapropane (bebt) and two of its silver(I) complexes are presented. The DNA-binding properties of the complexes were investigated.

2. Experimental

2.1. Materials, instrumentation, and physical measurements

Calf thymus DNA (CT-DNA) and ethidium bromide (EB) were obtained from Sigma–Aldrich. Tris–HCl buffer, $\text{Na}_2\text{HPO}_4\text{--NaH}_2\text{PO}_4$ buffer, and EDTA–Fe(II) solution were prepared using bidistilled water. Carboxylate silver was prepared by silver nitrate and sodium carboxylate reaction. Other reagents and solvents were reagent grade obtained from commercial sources and used without purification. The stock solution of the complex was dissolved in DMF at 3×10^{-3} M. The stock solution of DNA (2.5×10^{-3} M) was prepared in 5 mM Tris–HCl/50 mM NaCl buffer (pH 7.2, stored at 4 °C and used within 4 days). A solution of CT-DNA gave a ratio of UV absorbance at 260 and 280 nm of about 1.8–1.9, indicating that the CT-DNA was sufficiently free of protein [22].

C, H, and N elemental analyses were determined using a Carlo Erba 1106 elemental analyzer. IR spectra were recorded on a Nicolet FT-VERTEX 70 spectrometer from 4000 to 400 cm^{-1} using KBr pellets. Fluorescence spectra were recorded on a LS-45 spectrofluorophotometer. Electronic spectra were taken on a Lab-Tech UV Bluestar spectrophotometer. Electrolytic conductance measurements were made with a DDS-11A type conductivity bridge using 10^{-3} mol dm^{-3} solution in DMF at room temperature. ^1H NMR spectra were obtained with a Mercury plus 400 MHz NMR spectrometer with TMS as internal standard and CDCl_3 as solvent.

2.2. DNA-binding studies

To compare quantitatively the affinity of the Ag(I) complexes toward DNA, the intrinsic binding constants K_b to CT-DNA were determined by monitoring the changes in absorbance with increasing concentration of DNA. From the absorption titration data, the binding constant was determined using the following equation [23]:

$$[\text{DNA}]/(\epsilon_a - \epsilon_f) = [\text{DNA}]/(\epsilon_b - \epsilon_f) + 1/K_b(\epsilon_b - \epsilon_f)$$

where [DNA] is the concentration of CT-DNA in the base pairs, ϵ_a corresponds to the observed extinction coefficient ($A_{\text{obsd}}/[M]$), ϵ_f corresponds to the extinction coefficient of the free compound, ϵ_b is the

extinction coefficient of the compound when fully bound to CT-DNA, and K_b is the intrinsic binding constant. The ratio of slope to intercept in the plot of $[\text{DNA}]/(\varepsilon_a - \varepsilon_f)$ versus $[\text{DNA}]$ gave the value of K_b . EB emits intense fluorescence in the presence of CT-DNA due to its strong intercalation between the adjacent CT-DNA base pairs. It was previously reported that the enhanced fluorescence can be quenched by the addition of a second molecule [24, 25]. The extent of fluorescence quenching of EB bound to CT-DNA can be used to determine the extent of binding between the second molecule and CT-DNA. Competitive binding experiments were carried out in the buffer by keeping $[\text{DNA}]/[\text{EB}] = 1.13$ and varying the concentrations of the compounds. The spectra were analyzed according to the classical Stern–Volmer equation [26]:

$$I_0/I = 1 + K_{sv}[Q]$$

where I_0 and I are the fluorescence intensities at 599 nm in the absence and presence of the quencher, respectively, K_{sv} is the linear Stern–Volmer quenching constant, and $[Q]$ is the concentration of the quencher. In these experiments, $[\text{CT-DNA}] = 2.5 \times 10^{-3} \text{ mol L}^{-1}$, $[\text{EB}] = 2.2 \times 10^{-3} \text{ mol L}^{-1}$.

Viscosity experiments were conducted on an Ubbelohde viscometer, immersed in a water bath maintained at $25.0 \pm 0.1^\circ\text{C}$. Titrations were performed for the complex (1–10 μM), and each compound was introduced into CT-DNA solution (50 μM) present in the viscometer. Viscosity values were calculated from the observed flow times of CT-DNA containing solutions corrected for the flow time of buffer alone (t_0), $\eta = (t - t_0)$ [27]. Data were presented as $(\eta/\eta_0)^{1/3}$ versus the ratio of the concentration of the complex to CT-DNA, where η is the viscosity of CT-DNA in the presence of the complex and η_0 is the viscosity of CT-DNA alone.

2.3. Synthesis of ligand and complexes

2.3.1. Preparation of 1,3-bis(1-ethylbenzimidazol-2-yl)-2-thiapropane (bebt)

1,3-bis(benzimidazol-2-yl)-2-thiapropane (5.88 g, 0.02 mol), synthesized by a literature method [28–31], was suspended in dry tetrahydrofuran (150 mL) and stirred under reflux with potassium (1.56 g, 0.04 mol). The solution was stirred until potassium disappeared, then iodoethane (7.70 g, 0.05 mol) was added. After 1 h, the solvent was concentrated and the resulting powder was washed with distilled water several times to remove soluble KI. The undissolved substances were recrystallized with methanol and pale-yellow powder of ligand was deposited. Yield 5.50 g (78.6%); m.p. 170–173 $^\circ\text{C}$. Anal. $\text{C}_{20}\text{H}_{22}\text{N}_4\text{S}$: Calcd C 67.61; H 6.19; N 15.77%; found: C 67.78; H 6.03; N 15.63%. ^1H NMR (400 MHz, CDCl_3) δ : 7.31 (m, 4H, Ph), 4.12 (m, 2H, CH_2), 4.28 (s, 2H, SCH_2), 1.49 (m, 3H, CH_3). IR (selected data, KBr): $\nu^- = 1274$ ($\nu_{\text{C-N}}$), 1458 ($\nu_{\text{C=N}}$), 1616 ($\nu_{\text{C=C}}$) cm^{-1} . UV/Vis (DMF): $\lambda = 270, 280$. Λ_M (DMF, 297 K): $28.9 \text{ S cm}^2 \text{ mol}^{-1}$.

2.3.2. $[\text{Ag}(\text{bebt})(\text{methacrylate})] (1)$

To a stirred solution of bebt (0.035 g, 0.10 mmol) in EtOH (5 mL) was added silver methacrylate (0.0193 g, 0.1 mmol). The resultant solution was allowed to slowly evaporate at room temperature in the dark. The block crystals suitable for X-ray diffraction studies were obtained after 2 weeks. Yield 0.042 g (77.3%). $\text{C}_{24}\text{H}_{27}\text{AgN}_4\text{O}_2\text{S}$: Calcd C, 52.99; H, 4.97; N, 10.30%; found: C, 53.04; H, 5.01; N, 10.31%. IR (selected data, KBr): $\nu^- = 1180$ ($\nu_{\text{C-N}}$), 1430 ($\nu_{\text{C=N}}$), 1655 ($\nu_{\text{C=C}}$), 1560 (ν_{asCOO}), 1430 (ν_{sCOO}) cm^{-1} . UV/Vis (DMF): $\lambda = 269, 282 \text{ nm}$. Λ_M (DMF, 297 K): $4.9 \text{ S cm}^2 \text{ mol}^{-1}$.

2.3.3. $[\text{Ag}_2(\text{bebt})_2(\text{pic})_2] (2)$

To a stirred solution of 1,3-bis(1-ethylbenzimidazol-2-yl)-2-thiapropane (0.175 g, 0.50 mmol) in hot MeOH (5 mL) was added Ag(I) picrate (0.1680 g, 0.50 mmol). A yellow precipitate formed rapidly in the solution. The precipitate was filtered off, washed with MeOH and absolute Et_2O , and dried *in vacuo*. The dried precipitate was dissolved in DMF resulting in a yellow solution. Yellow crystals suitable for X-ray diffraction studies were obtained by ether diffusion into DMF after several days at room temperature. Yield 0.29 g (84.5%). $\text{C}_{52}\text{H}_{48}\text{Ag}_2\text{N}_{14}\text{O}_{14}\text{S}_2$: Calcd C, 45.49; H, 3.52; N, 14.28%; found: C, 45.45; H, 3.54; N,

14.27%. IR (selected data, KBr): $\nu^- = 1265 (\nu_{\text{C-N}})$, $1434 (\nu_{\text{C=N}})$, $1633 (\nu_{\text{C=C}})$ cm^{-1} . UV/Vis (DMF): $\lambda = 350$ nm. Λ_{M} (DMF, 297 K): $78.9 \text{ S cm}^2 \text{ mol}^{-1}$.

2.4. X-ray crystal structure determination

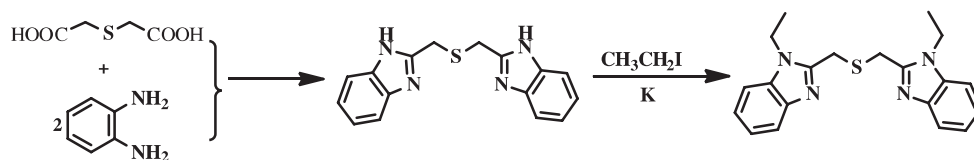
A suitable single crystal was mounted on a glass fiber and the intensity data were collected with a Bruker APEX II area detector with graphite-monochromated Mo-K α radiation ($\lambda = 0.71073$ Å) at 296 K. Data reduction and cell refinement were performed using SAINT [32]. The absorption corrections were made by empirical methods. The structures were solved by direct methods and refined by full-matrix least-squares against F^2 using SHELXTL software [33]. The non-H atoms in the structure were subjected to anisotropic refinement. Hydrogens were located geometrically and treated with the riding model. All hydrogens were found in different electron maps and subsequently refined in a riding model approximation with C–H distances ranging from 0.93 to 0.97 Å. The crystal data and experimental parameters relevant to the structure determination are listed in table 1. Selected bond distances and angles are shown in table 2.

Table 1. Crystal data and structure refinement for **1** and **2**.

Formula	1	2
Molecular formula	$\text{C}_{24}\text{H}_{27}\text{AgN}_4\text{O}_2\text{S}$	$\text{C}_{52}\text{H}_{48}\text{Ag}_2\text{N}_{14}\text{O}_{14}\text{S}_2$
Molecular weight	543.43	1372.90
Crystal system	Triclinic	Monoclinic
Space group	P-1	C2/c
a (Å)	10.245(4)	17.1358(12)
b (Å)	12.390(5)	23.2259(16)
c (Å)	13.110(5)	13.8863(9)
α (°)	92.911(5)	90.00
β (°)	112.323(4)	94.8780(10)
γ (°)	112.978(4)	90.00
V (Å ³)	1378.0(9)	5506.7(6)
Z	2	4
ρ_{calcd} (g cm ⁻³)	1.310	1.656
$F(000)$	556	2784
Crystal size (mm)	$0.35 \times 0.29 \times 0.23$	$0.36 \times 0.34 \times 0.30$
θ min/max (°)	1.73 to 25.50	1.48 to 26.00
$h/k/l$ (max, min)	–12, 12/–15, 14/–15, 15	–21, 19/–26, 28/–17, 16
Reflections collected	9324	15,173
Independent reflections	5013 [$R(\text{int}) = 0.0330$]	5397 [$R(\text{int}) = 0.0243$]
Refinement method	Full-matrix least-squares on F^2	Full-matrix least-squares on F^2
Data/restraints/parameters	5013/128/292	5397/6/381
Goodness-of-fit on F^2	1.100	1.029
Final R_1 , wR_2 indices [$I > 2\sigma(I)$]	0.0641, 0.1981	0.0456, 0.1193
R_1 , wR_2 indices (all data)	0.0777, 0.2120	0.0642, 0.1333
Largest diff. peak and hole (e Å ⁻³)	1.964 and –0.756	0.404 and –0.523

Table 2. Selected bond lengths (Å) and angles (°) of **1** and **2**.

1			
Ag(1)–N(2)	2.255(5)	N(2)–Ag(1)–N(3)	108.80(15)
Ag(1)–N(3)	2.300(4)	N(2)–Ag(1)–O(2)	142.05(16)
Ag(1)–O(2)	2.426(5)	N(3)–Ag(1)–O(2)	109.04(16)
2			
Ag(1)–N(1)#1	2.129(3)	N(1)#1–Ag(1)–N(3)	157.38(13)
Ag(1)–S(1)	2.7672(11)	N(1)#1–Ag(1)–S(1)	123.76(9)
Ag(1)–N(3)	2.163(3)	N(3)–Ag(1)–S(1)	76.97(9)



Scheme 1. Synthesis of bebt.

3. Results and discussion

3.1. Characterization of the complexes

The synthetic route to free bebt is shown in scheme 1. The Ag(I) complexes are soluble in polar aprotic solvents such as DMF and DMSO, partially soluble in water, ethanol, and methanol, but insoluble in Et₂O and petroleum ether. The results of the elemental analyses show compositions of [Ag(bebt)(methacrylate)] and [Ag₂(bebt)₂](pic)₂, which were confirmed by crystal structure analysis. The molar conductance values show that **2** is a 1 : 2 electrolyte, while **1** is a non-electrolyte in DMF [34].

The IR spectrum of bebt shows characteristic absorption bands at 1274 and 1458 cm⁻¹, which can be assigned to $\nu_{(\text{C}=\text{N})}$ and $\nu_{(\text{C}=\text{N})}$, respectively [30, 31]. The frequencies of **1** and **2** are also observed, but remarkably shifted by 10–100 cm⁻¹ for the two complexes. Since the carboxylate group can coordinate to the metal ion in a bidentate or a monodentate fashion, the “ Δ criterion,” which is based on the difference between the $\nu_{\text{as}}(\text{O}=\text{C}=\text{O})$ and $\nu_{\text{s}}(\text{O}=\text{C}=\text{O})$ values, compared to the corresponding value in sodium carboxylate, indicates the coordinating mode of carboxylate. The data suggest that carboxylate groups in **1** are bidentate ligands. This is confirmed by the result of the crystal structure analysis [35–38]. These absorptions support the argument that the nitrogens of the ligand are coordinated. The electronic spectra of bebt and the silver complexes were recorded in DMF solution at room temperature. The UV band of bebt (270, 280) is marginally shifted in **1**, assigned to $\pi \rightarrow \pi^*$ (benzene) and $\pi \rightarrow \pi^*$ (imidazole) transitions, which shows clear evidence of (C=N–C=C) coordination to silver [39]. For **2**, the UV band of 350 nm is assigned to picrate anion.

3.2. Description of the structures of **1** and **2**

Complex **1** crystallizes in the triclinic space group P-1 and its structure along with the atom numbering scheme is shown in figure 1. Complex **1** has a mononuclear four-coordinate configuration, consisting of Ag(I), plus bebt and methacrylate ligands. The Ag(I) is coordinated by two oxygens from a methacrylate anion and two nitrogens from a bebt. The structure of **1** is a distorted tetrahedral geometry. The methacrylate group shows coordination to Ag(I) in the chelated mode with the distances of Ag–O(1) 2.504(7) and Ag–O(2) 2.426(5) Å.

The X-ray crystallographic analysis reveals that **2** crystallizes in monoclinic space group C2/c and its structure along with the atom numbering scheme is shown in figure 2. Complex **2** can be described as a 2 + 1 coordinated silver(I) compound with a nearly linear N–Ag–N backbone and a Ag–S interaction. The sum of van der Waals radii of Ag–S is 3.57, so it is shorter than van der Waals radii as shown in table 2. The asymmetric unit of **2** consists of a binuclear dicationic [Ag₂(bebt)₂]²⁺ and two picrate anions. Two ligands are arranged in a face-to-face syn conformation to coordinate to two Ag(I) ions from opposite directions, generating a U-type double layer geometry around the metal ions.

As one important supramolecular force, π – π stacking shows a specific structural requirement for substrate recognition or the arrangement of complicated architectures. In **1**, as shown in figure 3(a), the centroid-to-centroid distances are 3.415 Å, indicating strong intermolecular interactions. In **2**, as shown in figure 3(b), the distances are 3.612 and 3.716 Å, respectively, indicating strong intramolecular π – π stacking.

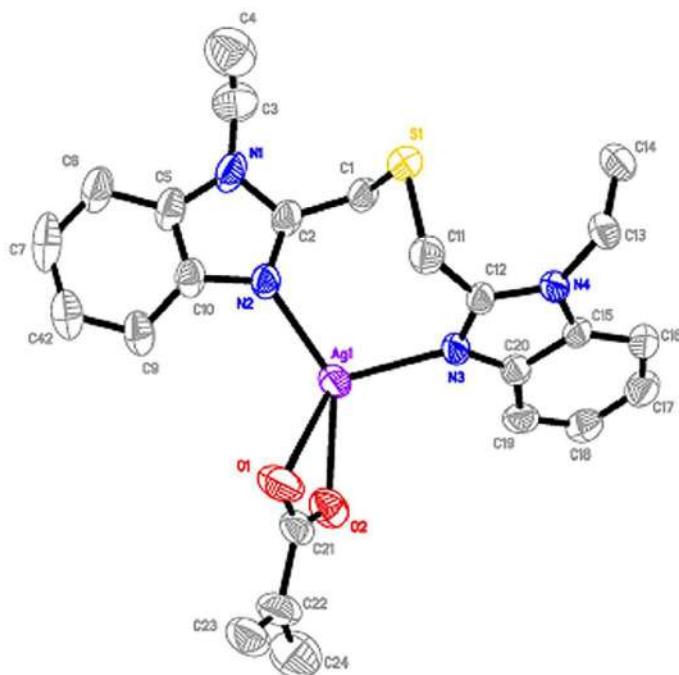


Figure 1. Molecular structure of $[Ag(bebt)(methacrylate)]$ with displacement ellipsoids drawn at the 30% probability level; hydrogens are omitted for clarity.

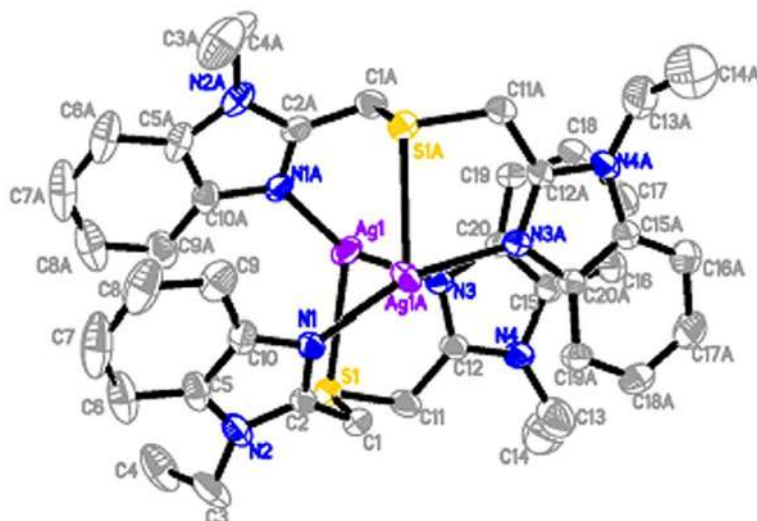


Figure 2. Molecular structure of $[Ag_2(bebt)_2]^{2+}$ with displacement ellipsoids drawn at the 30% probability level; anion and hydrogens are omitted for clarity.

3.3. DNA-binding properties

3.3.1. Absorption spectroscopic studies

We first investigated the binding of DNA with bebt and its complexes through absorption titration experiments. The electronic absorption spectra of free bebt and its Ag(I) complexes in the absence and

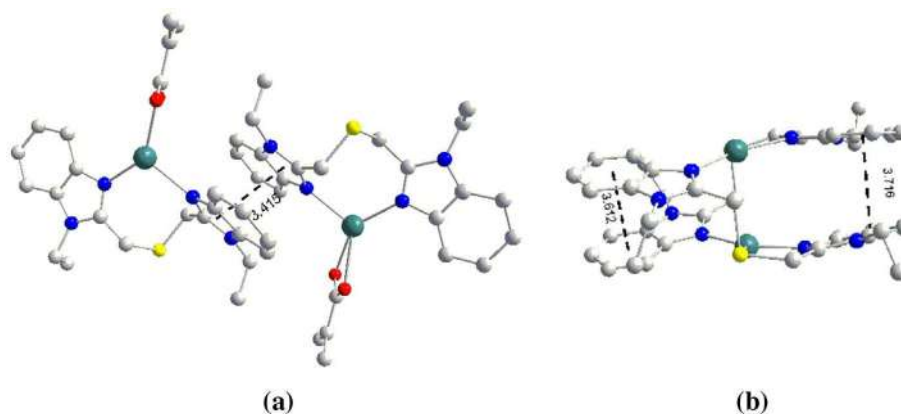


Figure 3. Structure of **1** (a) and **2** (b) linked by π - π stacking interactions.

presence of CT-DNA are given in figure 4. Upon addition of increasing CT-DNA concentrations, these bands exhibited hypochromism of 10.7, 41.7, and 9.6%, respectively. From the electronic absorption spectroscopy experiments, K_b values of free bebt, **1** and **2** were obtained as $1.22 \times 10^4 \text{ M}^{-1}$ ($R^2 = 0.9921$), $5.9 \times 10^4 \text{ M}^{-1}$ ($R^2 = 0.9958$) and $6.5 \times 10^4 \text{ M}^{-1}$ ($R^2 = 0.9931$), respectively, compared with those of so-called DNA intercalation complexes ($K_b = 2.9 \times 10^4 - 1.2 \times 10^5 \text{ M}^{-1}$) [40–44]. It may be suggested that the ligand and complexes can bind to DNA via intercalation and the order of the binding affinity is **2** > **1** > bebt.

In general, this phenomenon may be attributed to several factors. The charge transfer of coordinated bebt resulting from coordination to the Ag(I) atom should reduce the charge density of the benzimidazole, which is conducive to intercalation. So the affinity for DNA is stronger in case of **1** and **2** when compared with the ligand. The difference of the two Ag(I) complexes in DNA binding ability could be attributed to the presence of an electron deficient center in the charged complex **2** where an additional interaction between the complex and the phosphate-rich DNA backbone may take place. Also, comparison of their molecular structures shows a greater number of coplanar aromatic rings in **2**, which contributes to intercalation of the base pairs of double helical DNA, and which may result in stronger affinity for DNA.

3.3.2. Fluorescence spectroscopic studies

In order to examine the ability of the compounds to displace EB (ethidium bromide) from its EB–DNA complex, a competitive EB binding study was undertaken with UV/Vis and fluorescence experiments [45–49]. EB, a phenanthridine fluorescence dye, is a typical indicator of intercalation that forms soluble complexes with nucleic acids, emitting intense fluorescence in the presence of CT-DNA due to the intercalation of the planar phenanthridine ring between adjacent base pairs on the double helix [50]. The changes observed in the spectra of EB on its binding to CT-DNA are often used for study between DNA and other compounds such as metal complexes [51]. No luminescence was observed for the two complexes at room temperature in any organic solvent or in the presence of CT-DNA. Competitive EB binding studies could be undertaken to examine the binding of each complex with DNA. EB (ethidium bromide) is a conjugate planar molecule. If a complex can replace EB from DNA-bound EB, the fluorescence of the solution will be quenched due to the fact that free EB molecules are readily quenched by surrounding water molecules [52]. Stern–Volmer plots DNA–EB (figure 5) show that quenching of EB bound to DNA by the three compounds follows a linear relationship, consistent with intercalation of the test compounds. The K_{sv} values for free bebt, **1** and **2** were $2.49 \times 10^3 \text{ M}^{-1}$ ($R^2 = 0.9901$), $2.9 \times 10^3 \text{ M}^{-1}$ ($R^2 = 0.9973$) and $3.38 \times 10^3 \text{ M}^{-1}$ ($R^2 = 0.9957$), respectively. These observations suggest that the complexes can compete for DNA-binding sites with EB and displace EB from the EB–DNA system [53], characteristic of the intercalative interaction of compounds with DNA [54].

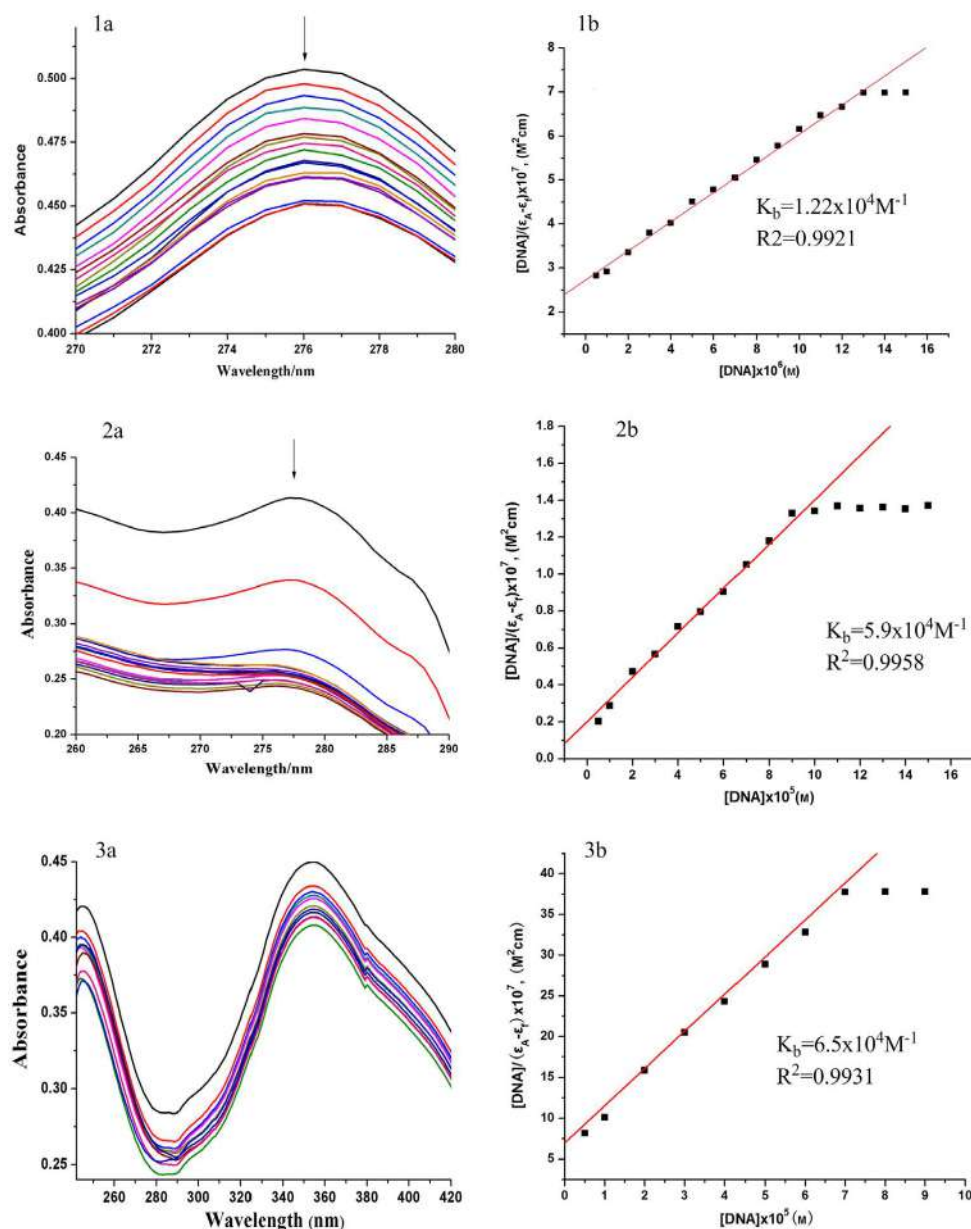


Figure 4. Electronic spectra of free bebt (1a), **1** (2a) and **2** (3a) in Tris-HCl buffer upon addition of CT-DNA. The arrow shows the emission intensity changes upon increasing DNA concentrations. [DNA]/(ε_a - ε_p) vs. [DNA] for the titration of the ligand bebt (1b), **1** (2b) and **2** (3b) with CT-DNA.

3.3.3. Viscosity measurements

Optical photophysical techniques are widely used to study the binding of ligand, metal complexes, and DNA, but do not give sufficient clues to support a binding model. Therefore, viscosity measurements were carried out to further clarify the interaction of **1** and **2** with DNA. Hydrodynamic measurements that are sensitive to the length change are regarded as the least ambiguous and most critical test of a binding model in solution in the absence of crystallographic structural data [55]. For bebt and the Ag(I) complexes, as increasing amounts of the compounds are added, the viscosity of DNA increases

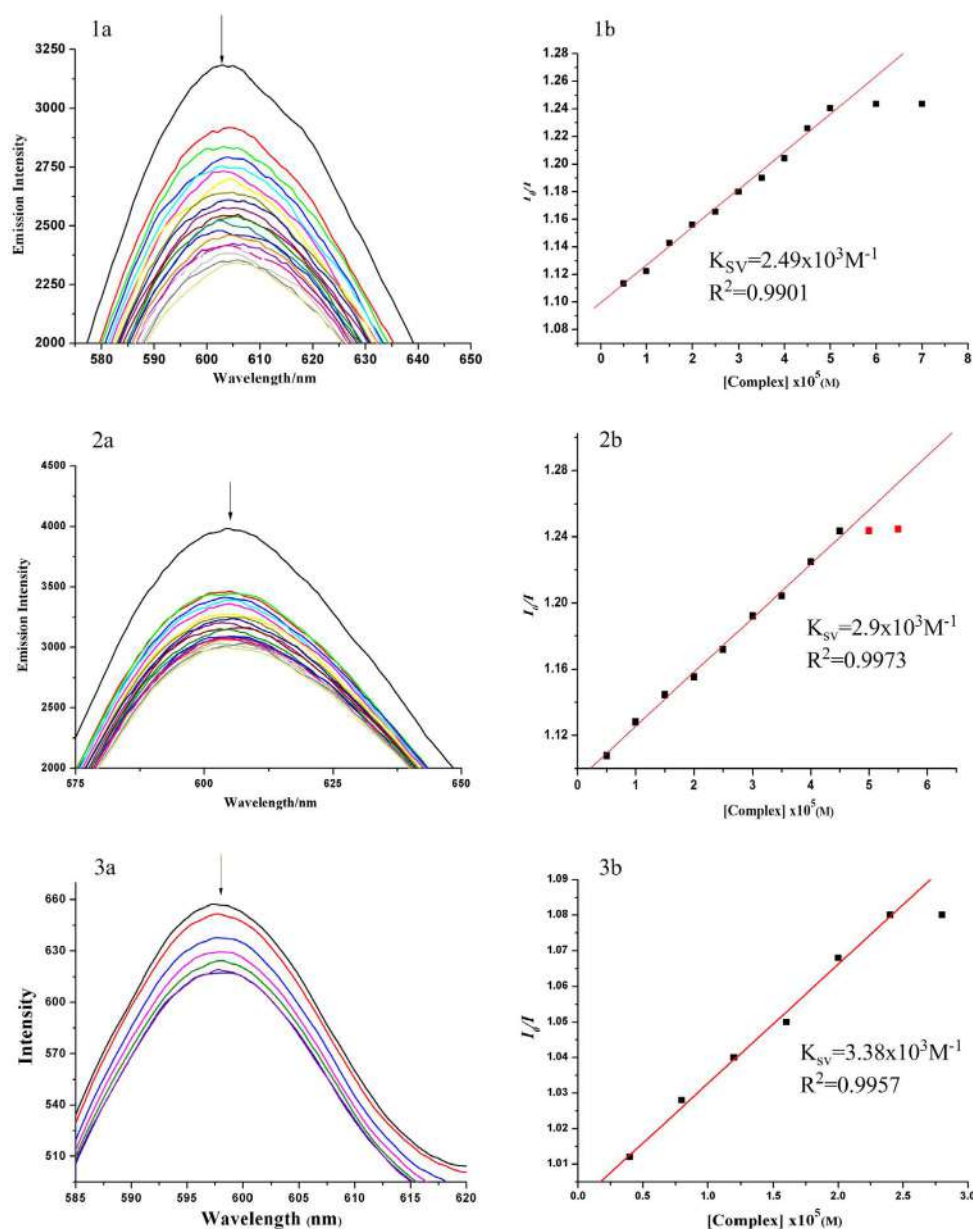


Figure 5. Emission spectra of EB bound to CT-DNA in the presence of free bebt (1a), **1** (2a), **2** (3a) $\lambda_{\text{ex}} = 520 \text{ nm}$. The arrows show the intensity changes upon increasing concentrations of the complexes. Fluorescence quenching curves of EB bound to CT-DNA by free bebt (1b), **1** (2b) and **2** (3b) (Plots of I_0/I vs. [Complex]).

steadily. The values of $(\eta/\eta_0)^{1/3}$ were plotted against [compound]/[DNA]. Viscosity measurements were carried out on CT-DNA by varying the concentration of the complexes. Figure 6 shows the ligand and complexes increase the relative viscosity of DNA. This may be explained by the compounds intercalating the adjacent DNA base pairs, leading to an increase in the separation of base pairs at intercalation sites and thus, an increase in overall DNA contour length. So the results demonstrate that the ligand and its complexes could bind to DNA by intercalation.

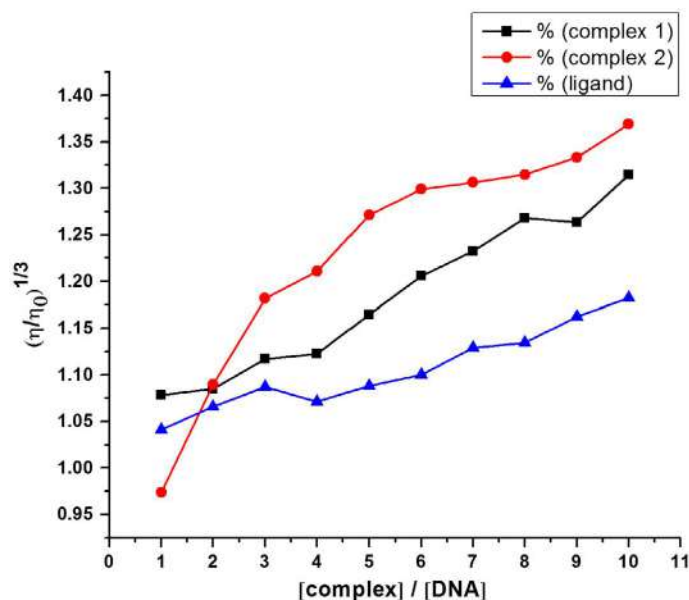


Figure 6. Effect of increasing amounts of the compounds on the relative viscosity at 25.0 ± 0.1 °C.

4. Conclusion

A new bis-benzimidazole ligand and its Ag(I) complexes were synthesized and structurally characterized. The interaction of the Ag(I) complexes with CT-DNA was explored using absorption and fluorescence spectral techniques and viscosity measurements. The data suggest that **1** and **2** bind to CT-DNA in an intercalation mode, and the complexes bind more strongly than the free ligand. Results obtained from this work would be useful to understand the mechanism of interactions of the small molecule binding to DNA and helpful in the development of their potential biological, pharmaceutical, and physiological implications [56–58].

Supplementary material

Crystallographic data (excluding structure factors) for the structure in this article have been deposited with the Cambridge Crystallographic Data Center as supplementary publication CCDC 1473233 and CCDC 1473234. Copies of the data can be obtained, free of charge, on application to the CCDC, 12 Union Road, Cambridge CB2 1EZ, UK (Fax: +44-1223-336-033; E-mail: deposit@ccdc.cam.ac.uk; or <http://www.ccdc.cam.ac.uk>).

Disclosure statement

No potential conflict of interest was reported by the authors.

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